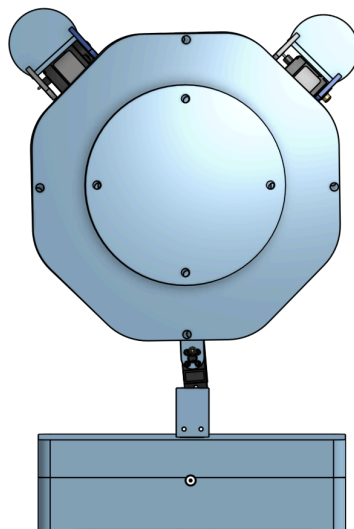


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August 15, 2026

Mech Challenge Documentation: Alarm Clock



Deliverables

CAD Model Link: [Onshape Design](#)

Design Overview & Decision Matrix

For this challenge, I wanted to turn the clock into an expressive little robot character that wakes you up using natural motion and personality before it makes some sound.

The character has these emotional states:

- **Sleeping:** The head slouches forward/down slightly, the ears droop, and the ambient LEDs slowly breathe with a dim, soft light.
- **Waking Up:** The head slowly lifts up and pans smoothly to face you, and the ears perk up.
- **Alarm Going Off:** The head looks straight at you, the ears flap back straight, and the alarm audio begins to play.
- **Snooze:** When you hit the snooze button on top, the head drops back into a slouch, the ears fall backward, and it goes back to resting.

How It Moves (6 Degrees of Freedom Total)

To make the character feel alive without making the wiring a total mess, I used FeeTech SCS0009 micro servos daisy-chained together to get 6 total DOFs:

- **The Head (2 DOFs):**
 - **Spin / Yaw:** The main base turns up to 330 degrees around so it can look left, right, forward, and almost all the way behind itself.
 - **Tilt / Nod (Pitch/Roll):** Tilts side-to-side and up-and-down to show curiosity or look like it's nodding.
- **The Ears (4 DOFs – 2 on each ear):**
 - **Left Ear:** Spins around 330 degrees (Yaw) and flaps front-to-back 180 degrees (Pitch).
 - **Right Ear:** Spins around 330 degrees (Yaw) and flaps front-to-back 180 degrees (Pitch).

Why I Chose This Design?

When I started designing, I thought through three different directions:

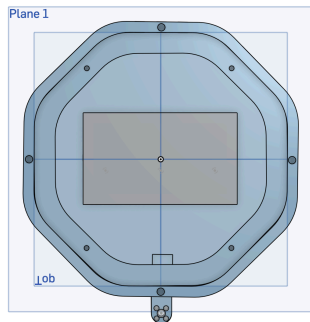
1. **Option 1: Just a Fixed Screen in a Box (Like an Echo Show)**
 - **Problem:** It would be super easy to 3D print and assemble, but it has zero character or physical life. It completely misses the point of the challenge.
2. **Option 2: A Screen with 1 Simple Nodding Motor**
 - *The Problem:* Adding just one motor lets the screen tilt up and down, but that's really only one gesture (nodding). It still feels like a generic appliance with a hinge.

3. Option 3: Fully Articulated Head & Moving Ears (My Chosen Design)

- *Why it won:* Putting moving ears on top of a panning head gives you real character expressions (like a curious puppy or Mickey Mouse ears perking up). The main engineering challenge was making sure 6 motors did not turn into a huge mess along with the other wiring as well. By splitting the mechanical enclosure into 9 clean modular parts and designing a custom 2-layer PCB that hides all the motor connections on the back, I was able to get full 6-DOF movement while keeping the assembly clean and straightforward.

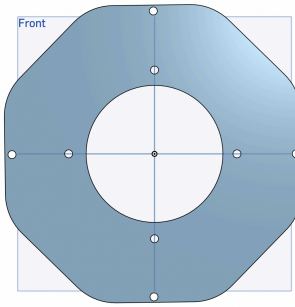
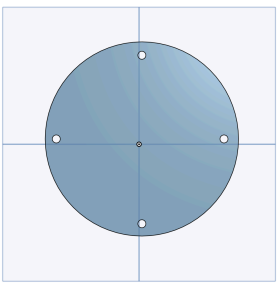
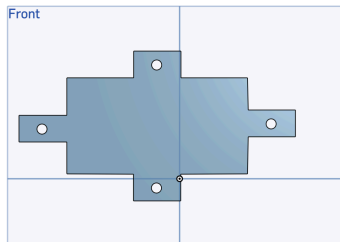
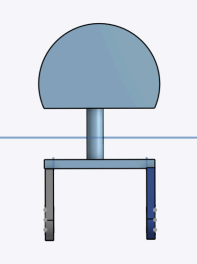
DFM & DFA (How it's Made & Assembled)

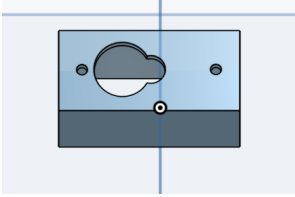
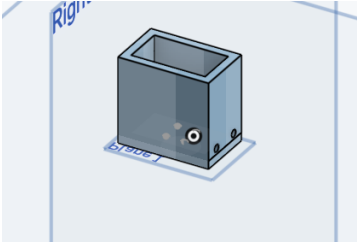
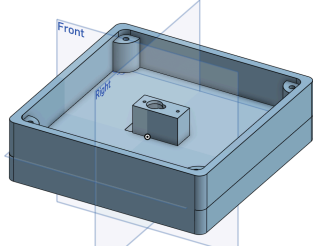
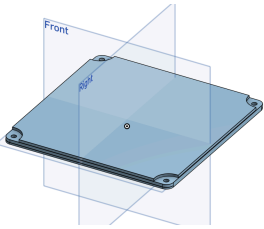
I broke everything into 9 3D printable parts so everything prints without any overhangs and can be put together easily like mentioned.



Clock Main Base

Sized at 6.5 inches wide, 4.63 inches deep, and 0.5 inches face thickness. I made it an octagon with filleted (rounded) corners so it prints cleanly without warping at sharp edges. It has built-in rectangular pass-through holes so the red and black power wires slide right through to the back of the PCB without getting pinched.

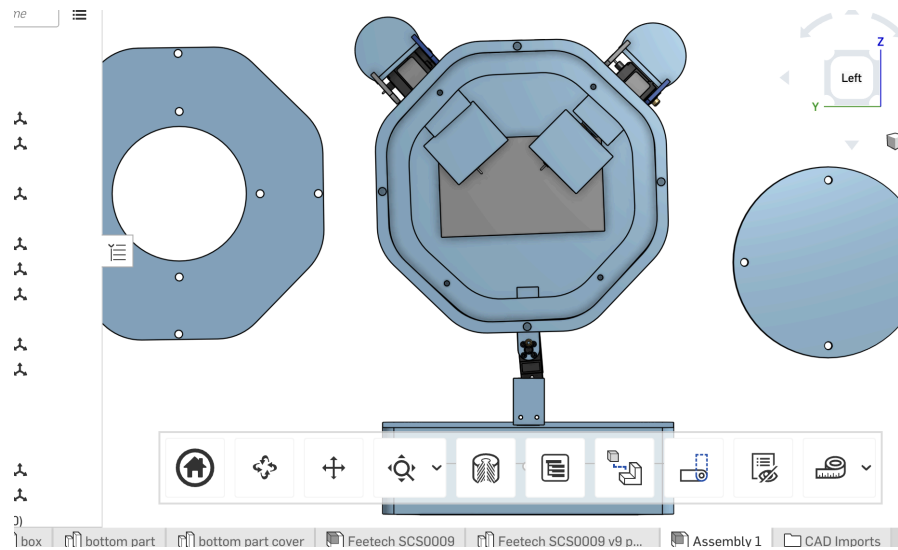
 Front Clock Cover Base	Flat front bezel with a circular cutout for the 3-inch display. Prints flat with zero support material and has alignment screw holes matching the main base.
 Clear Case (Screen Lens)	Printed out of transparent filament (Clear PETG) to protect the 3-inches LCD from dust and scratches. Drops right into the front cover bezel without needing messy adhesives.
 Front Battery Pack Cover	A symmetrical clamping plate that secures a standard 2x18650 battery sled inside the base using two quick screws.
 Two Ears (times 2)	Mickey-mouse shaped ear paddles with built-in mounting brackets. The arms attach directly to the micro servo horns so the ears can flap back and forth without extra linkage parts.

 Two Box servos (times 2)	Specially keyed cutout that holds the FeeTech SCS0009 servo body securely. This is for the vertical yaw for the ear.
 Three Box (times 3)	A simple rectangular housing that prints straight up without support. Also used to insert another servo in here as well. This is the vertical yaw servo can be attached to the servo that allows the two ears to flap back and forth and the head to go side to side.
 Bottom Part (Chassis)	Wide, sturdy bottom base that holds the main vertical yaw servo and keeps the center of mass low so the clock doesn't wobble when it turns.
 Bottom Part Cover	Flat plate that seals the bottom base. Screws on from the underside so you can easily access the battery and bottom servo without taking the entire head apart.

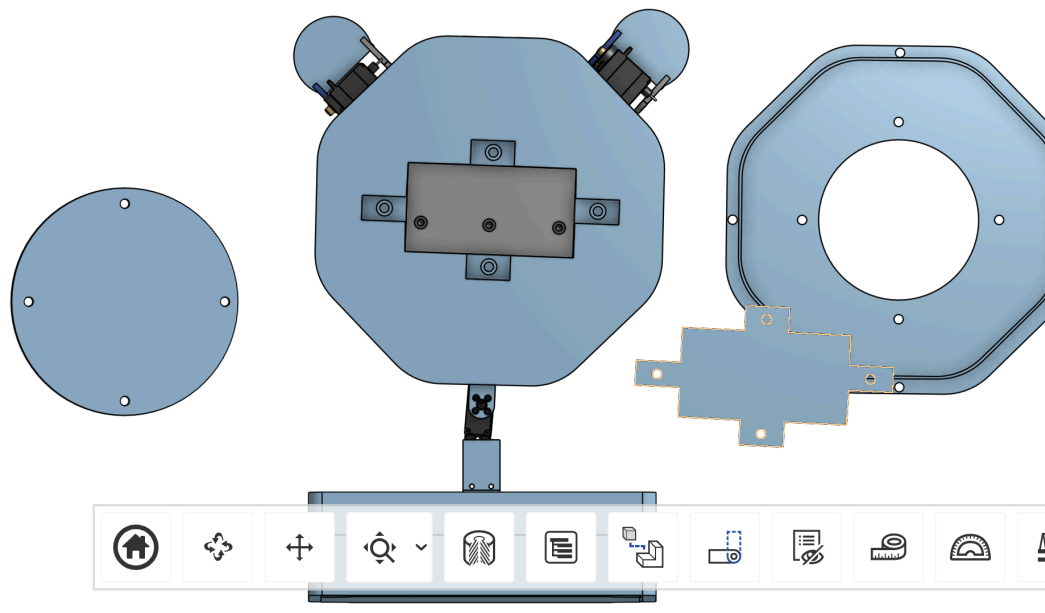
Hardware Standardization (DFA): I standardized all mounting hardware to use M2 machine screws 0.177 inches clearance pockets, 0.338 inches paired with M2 brass heat-set inserts melted into 5.0 mm deep standoffs. This means you only need one 1.5 hex driver to build the whole clock, and the brass threads won't strip if

you take it apart to tweak the electronics. This all needs to be further developed however. It would be more clear after some test prints as well the exact screw sizes.

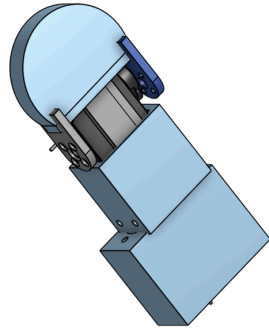
How the inside looks:



The backside:



One whole ear with servos, servo box, and box attached together:



Custom 2-Layer PCB Design

I put a custom 2-layer PCB inside the main base to separate the clean user controls on the front from the messy wiring and power circuitry on the back:

PCB layout:

Front Side (What User Sees)

- 3.0 Inches Circular display Screen
- 3 Tactile Push Buttons (Snooze, Alarm Set, Reset/Boot)
- WS2812B RGB LEDs for soft ambient lighting

Back Side (Internal Wiring)

- ESP32-S3 Microcontroller Module
- AMS1117 3.3V Voltage Regulator + TP4056 Battery Charger
- MAX98357A I2S Audio Amp for the speaker
- JST Connectors for daisy-chaining servos & battery pack
- 4.7k Ω Pull-Up Resistors + Power Filtering Capacitors
- Solid Copper Ground Plane to block motor electrical noise

I used 4.7k Ω pull-up resistors (tied to 3.3V) on the buttons and communication lines. When a button is pressed, it pulls the signal line to GND. This gives a reliable active-low trigger and prevents the ESP32 from glitching when the motors draw power.

Engineering Math & Calculations

We need to prove the FeeTech SCS0009 servo has enough torque to lift and tilt the head without stalling:

$$\tau = (m * g * r) * S$$

- Head Assembly Weight (m): Approximately 0.165 kg (Housing, LCD screen, PCB, hardware)
- Distance to Center of Mass (r): Approximately 4.8 cm/0.048 m
- Safety Factor (S) : 1.5

$$\tau = (0.165 \text{ kg} * 9.81 \text{ m/s}^2 * 0.048 \text{ m}) * 1.5 = 0.116 \text{ N} * \text{m} = 1.19 \text{ kg} * \text{cm}$$

Validation: The FeeTech SCS0009 provides 2.30 kg * cm of torque at 6V. Since we only need 1.19 kg*cm, we have a Factor of Safety of approximately 1.93, so the servo will run smooth and won't overheat.

B. Stress Check ("Snooze Slap" Test)

When someone is half-awake and smacks the top snooze button with around 30N of force which is approximately 6.7 lbs.

$$\sigma = (M*c)/I$$

- Maximum calculated bending stress on the pivot mounting boss is $\sigma = 14.2 \text{ MPa}$
- Standard 3D-printed PETG has a yield strength of approximately 45 MPa
- Factor of Safety: $45/14.2 = 3.17$. The plastic structure will easily take repeated morning hits without cracking.

C. Motor Vibration & Heat Management

- Vibration/Acoustics: Small servos can buzz and rattle plastic shells. I would add silicone damping rings between the servo mounts and the printed boxes (Part 7) to mechanically isolate motor chatter and keep the alarm speaker sounding clear.
- Thermal Dissipation: The ESP32 and voltage regulator only pull about 2.5W on average. The cable cutouts in the back provide

natural airflow to keep internal temperatures below 55 degrees Celsius without the need of noisy fans.

Material Choices

- Main Base, Covers & Chassis → PETG: Tougher and more heat-resistant than regular PLA, handles impacts from being slapped, and holds brass heat-set inserts securely without cracking.
- Clear Screen Cover (Part 3) → Transparent PETG / Acrylic: Clear and see-through to protect the LCD screen from dust and scratches.
- Joint Boxes & Brackets → Nylon / PETG: Low-friction and wear-resistant so rotating joints do not wear down or bind over time.
- Vibration Dampers → Molded Silicone Rubber: Soft rubber grommets that absorb motor buzzing and gear vibration.

Manufacturing Notes & Tolerances (GD&T)

When sending the Clock Main Base (Image 1) to a machine shop or manufacturer, put emphasis on surfaces matter the most and how accurate the cuts need to be:

- What it is made of & how it gets made:
 - For the working prototype, it gets 3D printed out of PETG.
 - For real mass production, it gets plastic injection molded out of ABS or PETG.
- The Main Flat Mounting Face (Datum A): The back mounting surface must be flat within 0.002 inches so the front cover and internal mounts sit completely flush without wobbling.
- The Center Servo Pivot Hole (Datum B): The main pivot hole needs a snug fit 0.472 inches with a +0.001 / -0.000 inches tolerance) so the servo spline slides in tight without slipping or binding.
- The M2 Screw Holes: The screw hole center positions have a true position tolerance of 0.003 inch relative to Datum A and Datum B so the screws always line up with the brass heat-set inserts.
- Main Face Thickness: The main front wall thickness is maintained at 0.500 inches +/- 0.005 inches (along with the overall 6.50

inches width and 3.63 inches depth) so the structural shell holds its shape without sinking or warping.

Product Launch Roadmap & Engineering Tasks

If I were leading the mechanical design from day one to the launch date, here is the timeline and the main engineering tasks (tickets) to get it built and shipped:

Phase 1: Concept & Prototyping (Weeks 1 to 4)

- Task 1: Finish the 3D model in CAD and check that the head and ears can turn without bumping into each other.
- Task 2: 3D print a quick test rig to test the micro servos under real weight and make sure they don't stall.
- Task 3: Size the custom 2-layer PCB layout so all the electronics fit cleanly inside the head.

Phase 2: Engineering Validation (EVT) (Weeks 5 to 12)

- Task 1: Add draft angles (slight slopes on the walls) and screw bosses into the CAD files so the parts can be injection molded.
- Task 2: Drop-test the 3D-printed body from a desk height of 1 meter onto wood to make sure the plastic doesn't snap.
- Task 3: Run the head back and forth continuously to verify that internal wires don't twist, rub, or wear out.

Phase 3: Design Validation (DVT) (Weeks 13 to 20)

- Task 1: Send the final 2D drawings to the factory to start cutting the steel injection molds.
- Task 2: Run long-term reliability tests on the ears and neck joints (moving them about or more than 100,000 times to check for wear).
- Task 3: Adjust the rubber dampening around the motors so the alarm speaker sounds clear without making the plastic shell rattle.

Phase 4: Production & Launch (PUT) (Weeks 21 to 26)

- Task 1: Inspect the first real batch of molded plastic parts from the factory to make sure dimensions match the drawings.

- Task 2: Set up the assembly line steps and write a simple assembly guide for the factory workers.
- Task 3: Run final packaging drop tests, box the units up, and launch the product.

Future Upgrades & Next Steps

Add a Center Slip-Ring: Right now the head turns 330 degrees so the wires don't wrap up. Adding a small 4-wire slip ring down the center shaft would let the head spin a full continuous 360 degrees.

Capacitive Touch Snooze: Instead of a mechanical push button on top, put a capacitive touch sensor on the inside of the top shell so you can tap anywhere on the head to snooze. This way a user can pet the alarm clock and it can create that more interactive feature.

Organic Motion Curves: Program smooth S-curve bezier acceleration in C++ on the ESP32 so the head and ears accelerate and slow down gently like a real living character just like the PIXAR lamp that was created as well.